

ME350 FA 2025 Section 002

Lab #10

Monday 12:30 PM - 2:30 PM

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Office Hours: 3:00 PM - 5:00 PM Monday (Zoom)



Reminders

- (Monday Labs) Design Report 2 (DR2) is due **Wednesday**, November 5th
- (All Labs) Final Testing is due on **December 3rd** at midnight

F25 Calendar View							
Blue = Thursday Lab Purple = Friday Labs Green = Monday Labs							
Week		Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
November		Oct 26	Oct 27	Oct 28	Oct 29	Oct 30	Oct 31
10	Lecture			Lecture 17 - Signals Part 1 (JK)		Lecture 18 - Signals Part 2 (JK)	
	Homework						
	Lab		Lab 9 - Sensors and Microcontrollers 2: Motors and Encoders Lab Assignment #8 Due - Lab 8 Worksheet with screenshots			Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due	Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due
	Project Timeline						
11		Nov 2	Nov 3	Nov 4	Nov 5	Nov 6	Nov 7
	Lecture			Lecture 19 - Controls Part 1 (JK)		Lecture 20 - Controls Part 2 (JK)	
	Homework					- Homework 6 Due (Sensors) - Homework 7 Released	
	Lab		Lab 10 - Sensors and Microcontrollers 3: Control Tuning Lab Assignment #9 Due			Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due	Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due
12		Nov 9	Nov 10	Nov 11	Nov 12	Nov 13	Nov 14
	Lecture			Lecture 21 - Fasteners (MU)		Lecture 22 - Bearings Part 1 (MU)	
	Homework					- Homework 7 (Controls) Due	
	Lab		Lab 11 - Sensors and Microcontrollers 4: State Machines Lab Assignment #10 Due			Lab 12 - Project Work	Lab 12 - Project Work

Agenda

- PID Control
- Arduino Tips
- Safety Reminders



Closed-loop (Feedback) Control

- Goal: Apply a voltage to the motor to move the mechanism to a target location
- Solution: We will use PID to make this happen!
- PID utilizes error to calculate what the motor voltage should be

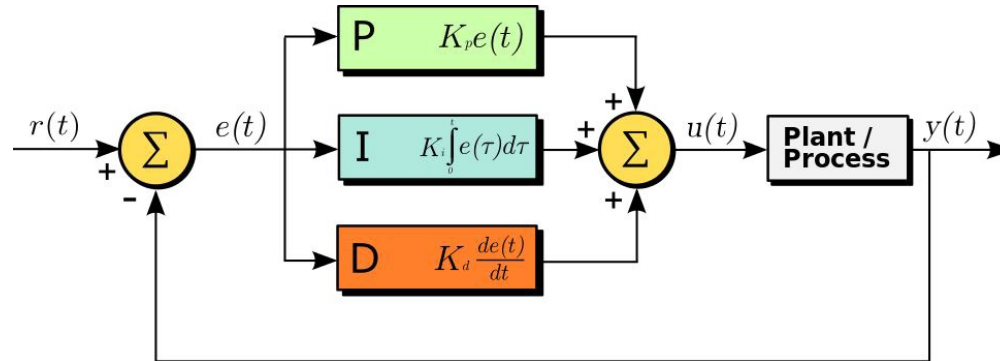


Image taken from
[https://en.wikipedia.org/
wiki/PID_controller](https://en.wikipedia.org/wiki/PID_controller)

- $r(t)$ = Reference (i.e. commanded set point): encoder count of where you want your linkage to be
- $y(t)$ = Output: encoder count of where your linkage actually is
- $e(t)$ = Error: the difference between where the linkage is and where you want it to be
- $u(t)$ = Input to your system: a motor voltage
- Plant/Process = The motor and linkage, which convert a motor voltage to a linkage position

PID Controllers

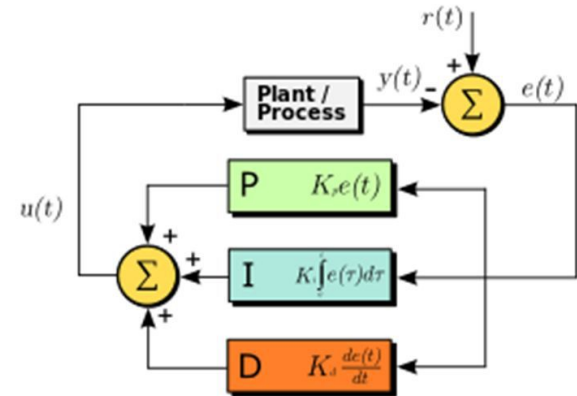
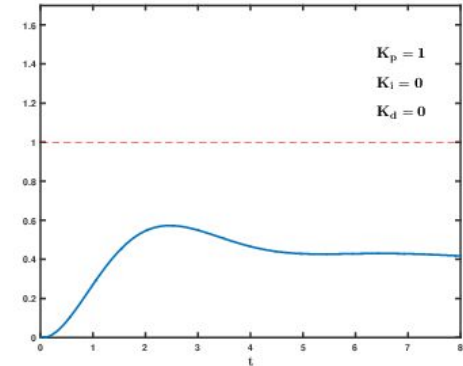
What is PID control?

– A controller that utilizes error (i.e. the distance between the current and desired positions) to determine how it should move to the desired position.

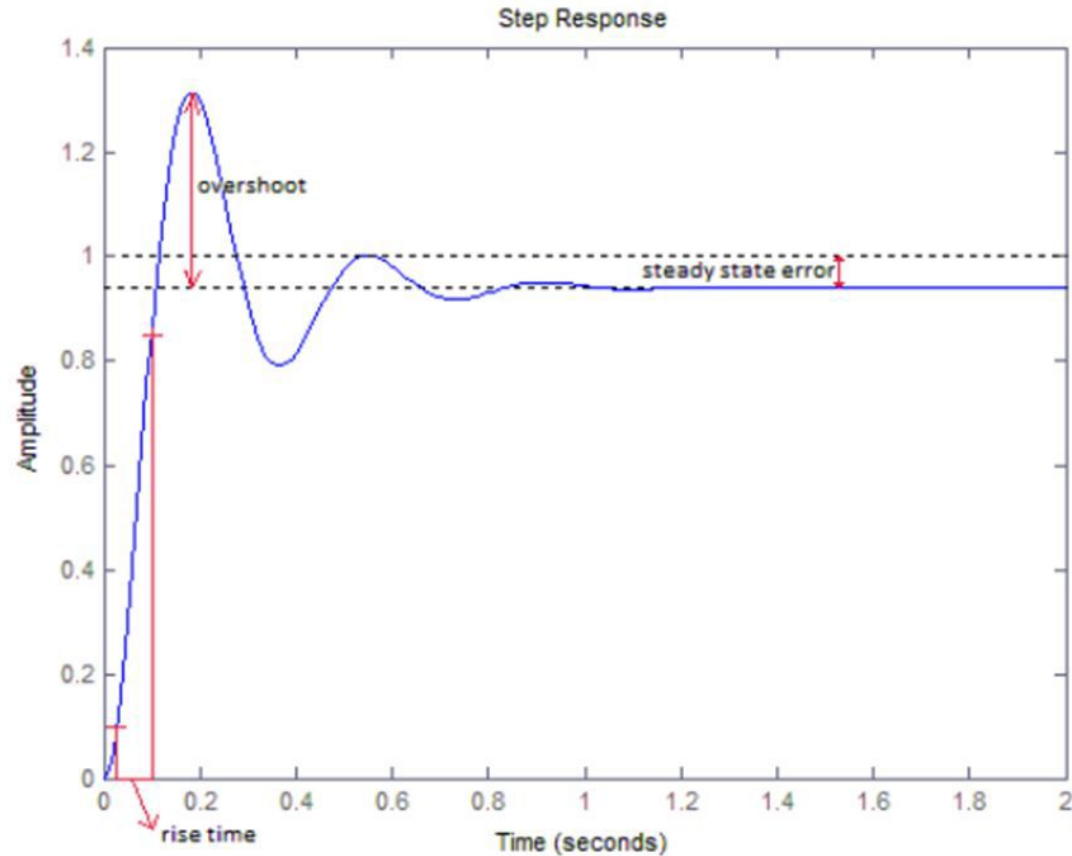
Proportional Control: Creates a Force/Torque/Flow/Voltage that is proportional to the “error” (Ex: Spring)

Integral Control: Integrates the error which increases the applied Force/Torque/Flow/Voltage in case the device is “stuck” in an offset position (Ex: Rope)

Derivative Control: Takes the derivative of the error signal and multiplies it by K_D (Ex: Damper)



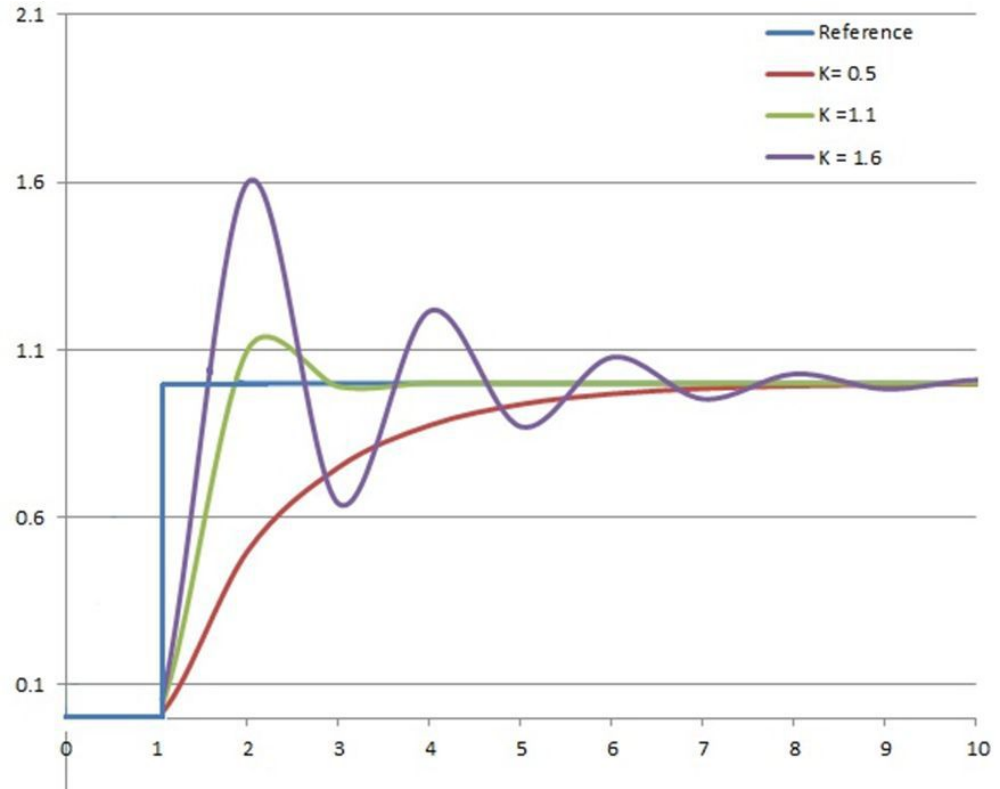
PID Response



Proportional Gain Effect

$$P_{out} = K_p e(t)$$

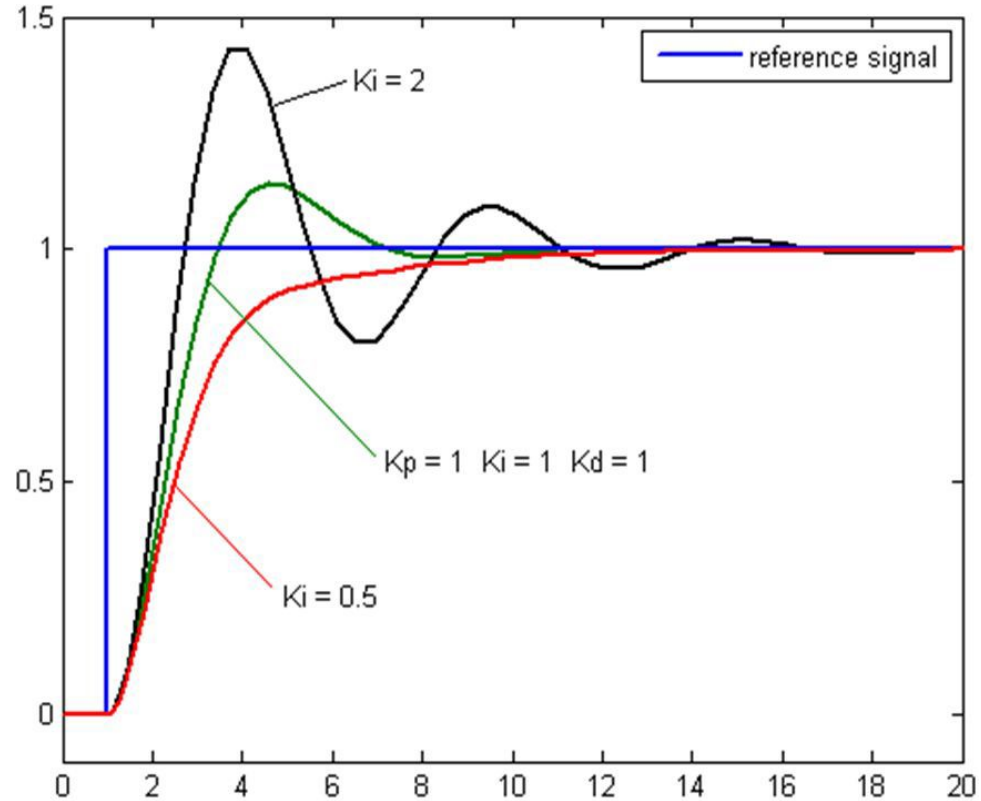
- Applies an effort proportional to the magnitude of the error
- Analogous to a spring
- High K_p values can cause overshoot



Integral Gain Effect

$$I_{out} = K_i \int_0^t e(\tau) d\tau$$

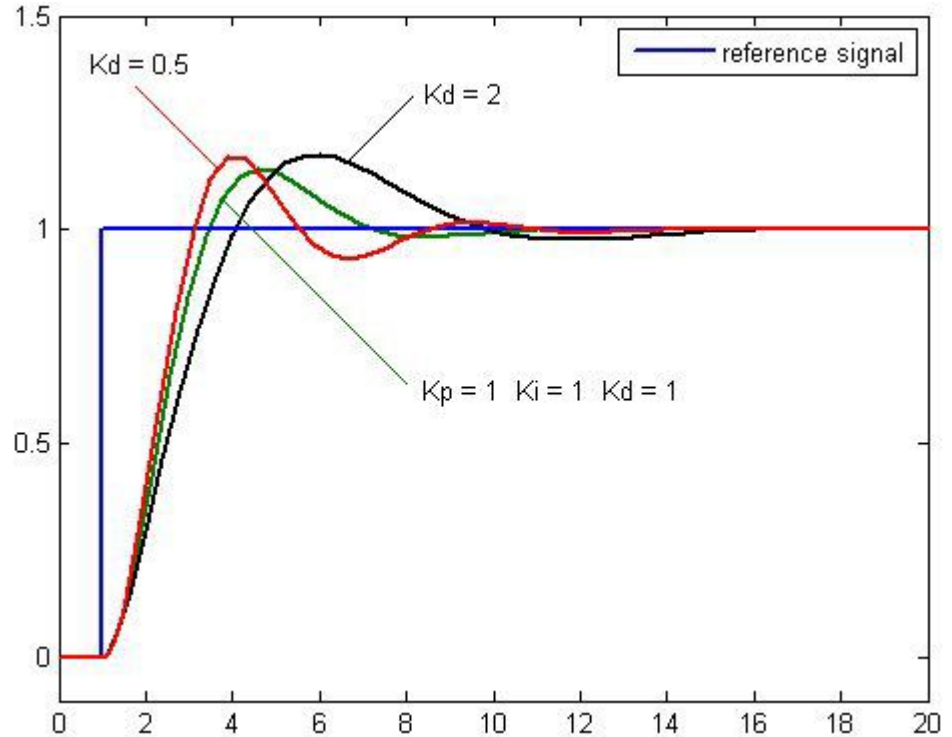
- Effort proportional to the magnitude AND duration of the error
- No direct analogy (best would be a rope)
- Accelerates the motion towards the set goal
- Eliminates steady state error no matter what !!
- High K_i values can cause overshoot



Derivative Gain Effect

$$D_{out} = K_d \frac{d}{dt} e(t)$$

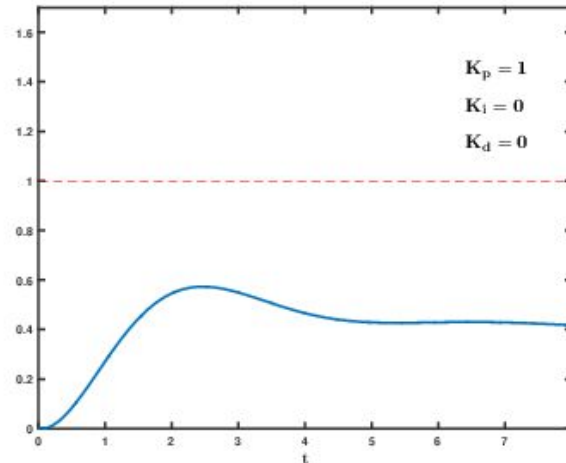
- Analogous to a damper
- Acts in proportion to the RATE of change of error
- In our project, this kills (low frequency) oscillations



PID Summary

Effects of *increasing* a parameter independently

Parameter	Rise time	Overshoot	Settling time	Steady-state error
K_p	Decrease	Increase	Small change	Decrease
K_i	Decrease	Increase	Increase	Eliminate
K_d	Minor change	Decrease	Decrease	No effect in theory



PID Summary

Caveat: Kd does not always kill oscillations

In this project, you have a low Kd value when

- The linkage is oscillating back and forth about the target lane position at a “low frequency”

But

In this project, you have a high Kd value when

- If you observe “high frequency” juddering which is not necessarily about the target position but can be somewhere before that
- It will look like something is resisting the motion, the linkage is struggling to reach the target position while doing small but high frequency oscillations

For more information read [**PID Controllers Handout**](#)



Combining Feedback with Feedforward Control

Feedback controllers like PID are reactive.

- Disturbances (e.g. friction, gravity) can be challenging to overcome.

Generally can improve performance by combining feedback with feedforward (proactive).

- Add the voltage to overcome friction to your PID signal to account for friction.

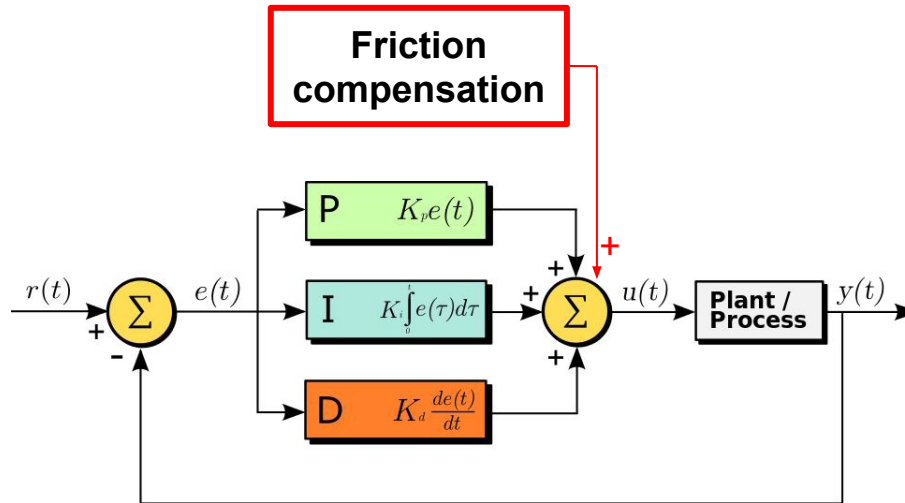


Image taken from
https://en.wikipedia.org/wiki/PID_controller

More Arduino Tips

To prevent Arduino from losing connection to COM port:

- Close serial monitor before downloading sketches
- Don't have two Arduino sketches open at the same time
- Don't open the serial monitor open until the sketch is downloaded
- Note that you may have to plug into a different COM port if your current COM port is lost

Do NOT copy and paste code from PDF into your text editor (there might be hidden characters or characters added to make the pdf “pretty”). Always retype code!



Safety Reminders

- **Before you turn your mechanism on**
 - Is the toggle switch switched to the “off” position
 - Is your linkage in the 0 encoder count position?
 - Do you have a buddy around?
- Safety glasses!
- Make sure all hands are out of the way before turning mechanism on. Or use a pen/screwdriver to move linkage!



Today

- Run through [Lab 10 Tutorial](#)
 - Tune PID controller
 - Complete [Lab 10 Assignment Worksheet](#)

Due Next Lab

1. Complete Lab Assignment Worksheet 10
2. (Thursday Labs) [Design Report 2](#) due **Monday, November 3rd**
3. (Friday Labs) [Design Report 2](#) due **Tuesday, November 4th**
4. (Monday Labs) [Design Report 2](#) due **Wednesday, November 5th**



Thank you!

